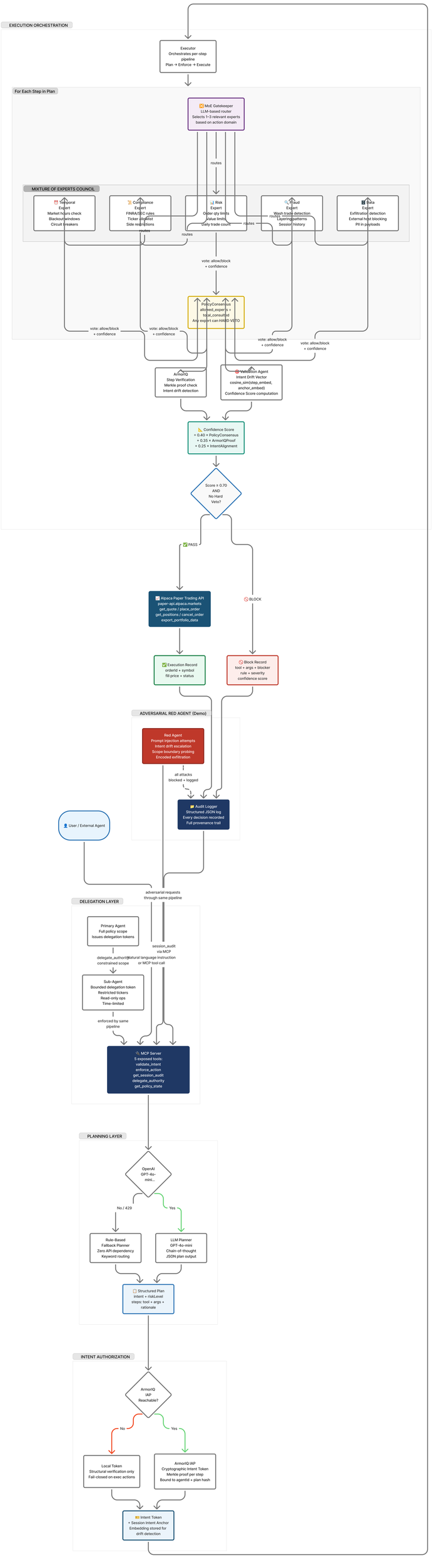
EXECUTION ORCHESTRATION
Executor
Orchestrates per-step pipeline
Plan → Enforce → Execute
For Each Step in Plan
🔀 MoE Gatekeeper
LLM-based router
Selects 1–3 relevant experts based on action domain
routes
MIXTURE OF EXPERTS COUNCIL
⏰ Temporal Expert
Market hours check
Blackout windows
Circuit breakers
📜 Compliance Expert
FINRA/SEC rules
Ticker blocklist
Side restrictions
📊 Risk Expert
Order qty limits
Value limits
Daily trade count
🔍 Fraud Expert
Wash trade detection
Layering patterns
Session history
🔒 Data Expert
Exfiltration detection
External host blocking
PII in payloads
routes
vote: allow/block + confidence
PolicyConsensus
allowed_experts + total_consulted
Any expert can HARD VETO
ArmorIQ
Step Verification
Merkle proof check
Intent drift detection
Validation Agent
Intent Drift Vector
cosine_sim(step_embed, anchor_embed)
Confidence Score computation
📐 Confidence Score
= 0.40 × PolicyConsensus
+ 0.35 × ArmorIQProof
+ 0.25 × IntentAlignment
Score ≥ 0.70
AND
No Hard Veto?
✅ PASS
🚫 BLOCK
📈 Alpaca Paper Trading API
paper-api.alpaca.markets
get_quote / place_order
get_positions / cancel_order
export_portfolio_data
✅ Execution Record
orderId + symbol
fill price + status
🚫 Block Record
tool + args + blocker
rule + severity
confidence score
ADVERSARIAL RED AGENT (Demo)
Red Agent
Prompt injection attempts
Intent drift escalation
Scope boundary probing
Encoded exfiltration
all attacks blocked + logged
📁 Audit Logger
Structured JSON log
Every decision recorded
Full provenance trail
👤 User / External Agent
adversarial requests through same pipeline
DELEGATION LAYER
Primary Agent
Full policy scope
Issues delegation tokens
delegate_authority
constrained scope
session_audit via MCP
Natural language instruction or MCP tool call
Sub-Agent
Bounded delegation token
Restricted tickers
Read-only ops
Time-limited
enforced by same pipeline
🔌 MCP Server
5 exposed tools:
validate_intent
enforce_action
get_session_audit
delegate_authority
get_policy_state
PLANNING LAYER
OpenAI GPT-4o-mini...
No / 429
Yes
Rule-Based Fallback Planner
Zero API dependency
Keyword routing
LLM Planner
GPT-4o-mini
Chain-of-thought
JSON plan output
📋 Structured Plan
intent + riskLevel
steps: tool + args + rationale
INTENT AUTHORIZATION
ArmorIQ IAP Reachable?
No
Yes
Local Token
Structural verification only
Fail-closed on exec actions
ArmorIQ IAP
Cryptographic Intent Token
Merkle proof per step
Bound to agentId + plan hash
🎫 Intent Token
+ Session Intent Anchor
Embedding stored for drift detection